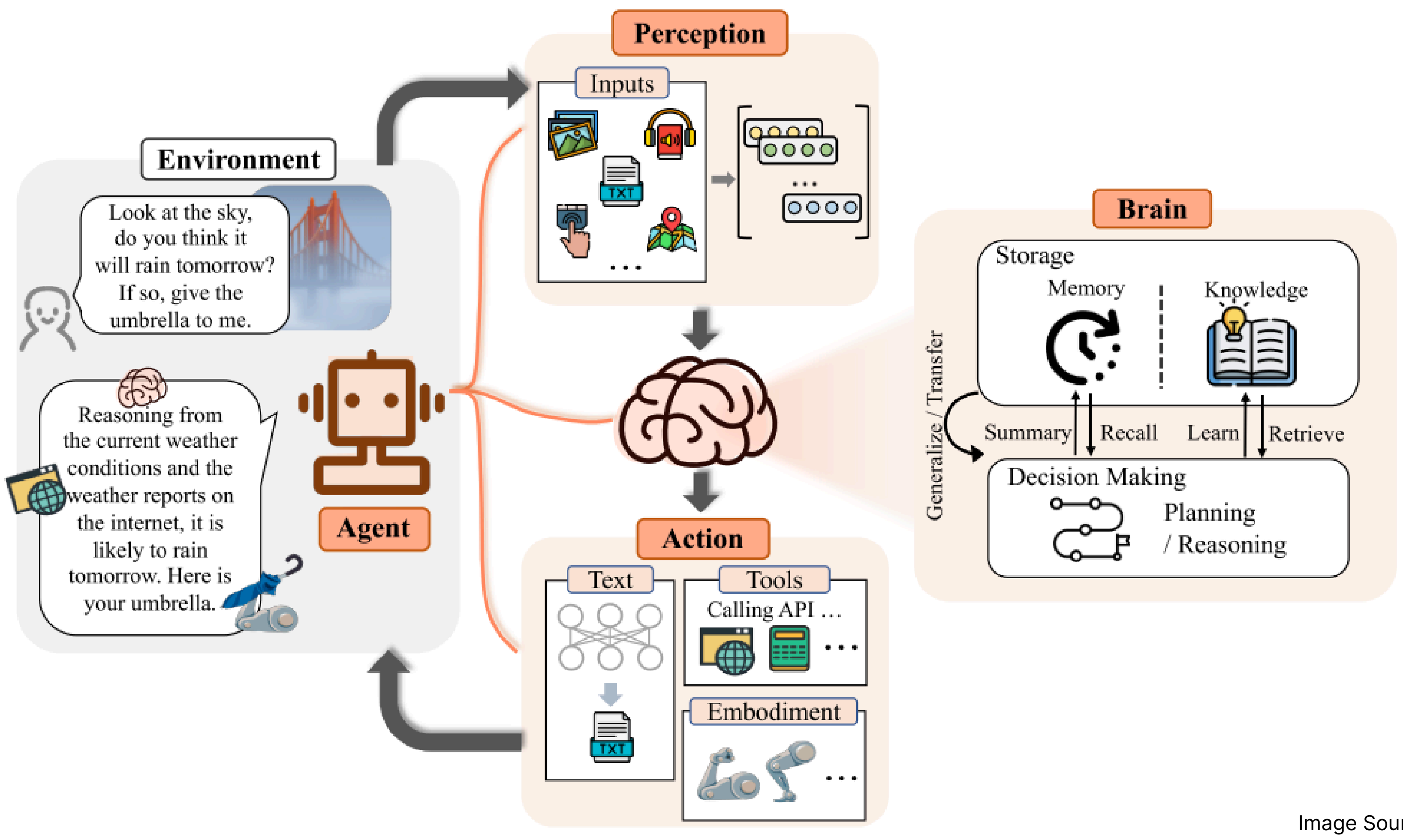
Day 7 of Agentic AI

How Agents Perceive: Sensors, Inputs & Multimodal Context



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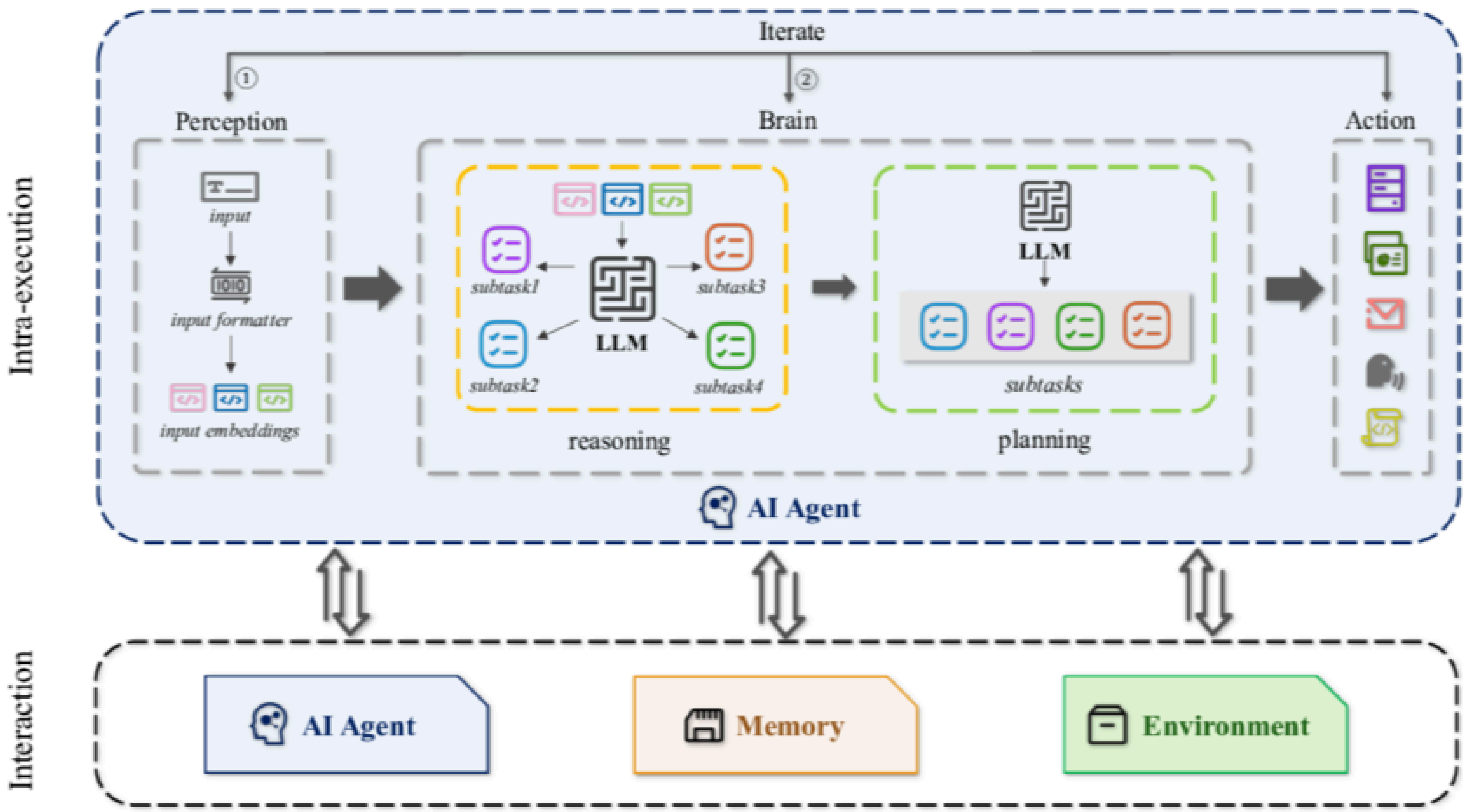


#30DaysofAgenticAI

Inputs: APIs, Databases, Sensors, and User Prompts

Agentic systems operate by continuously gathering information from a variety of inputs that represent both digital and physical contexts. These inputs form the foundation for the agent's perception, reasoning, and decision-making.

- **APIs:** The bridge to real-time data. Agents use APIs to fetch live information (like weather, stock prices, or user data) and take action across digital ecosystems.
- **Databases:** The access to persistent knowledge. From structured SQL to vector databases (like Pinecone), agents retrieve stored information, user profiles, logs, or semantic memories, to inform their decisions.
- **Sensors:** The link to the physical world. In IoT or robotics, sensors (cameras, mics, LiDAR) provide real-time environmental data that agents interpret to understand their surroundings.
- **User Prompts:** The human-in-the-loop. Direct inputs, through text, voice, or gesture, guide the agent, update its goals, and steer its reasoning on the fly.

Multimodal Perception

Multimodal perception allows agents to process and reason over multiple data types simultaneously text, images, audio, and video. This capability moves agents closer to human-like understanding, where meaning is derived from cross-referencing different sensory inputs.

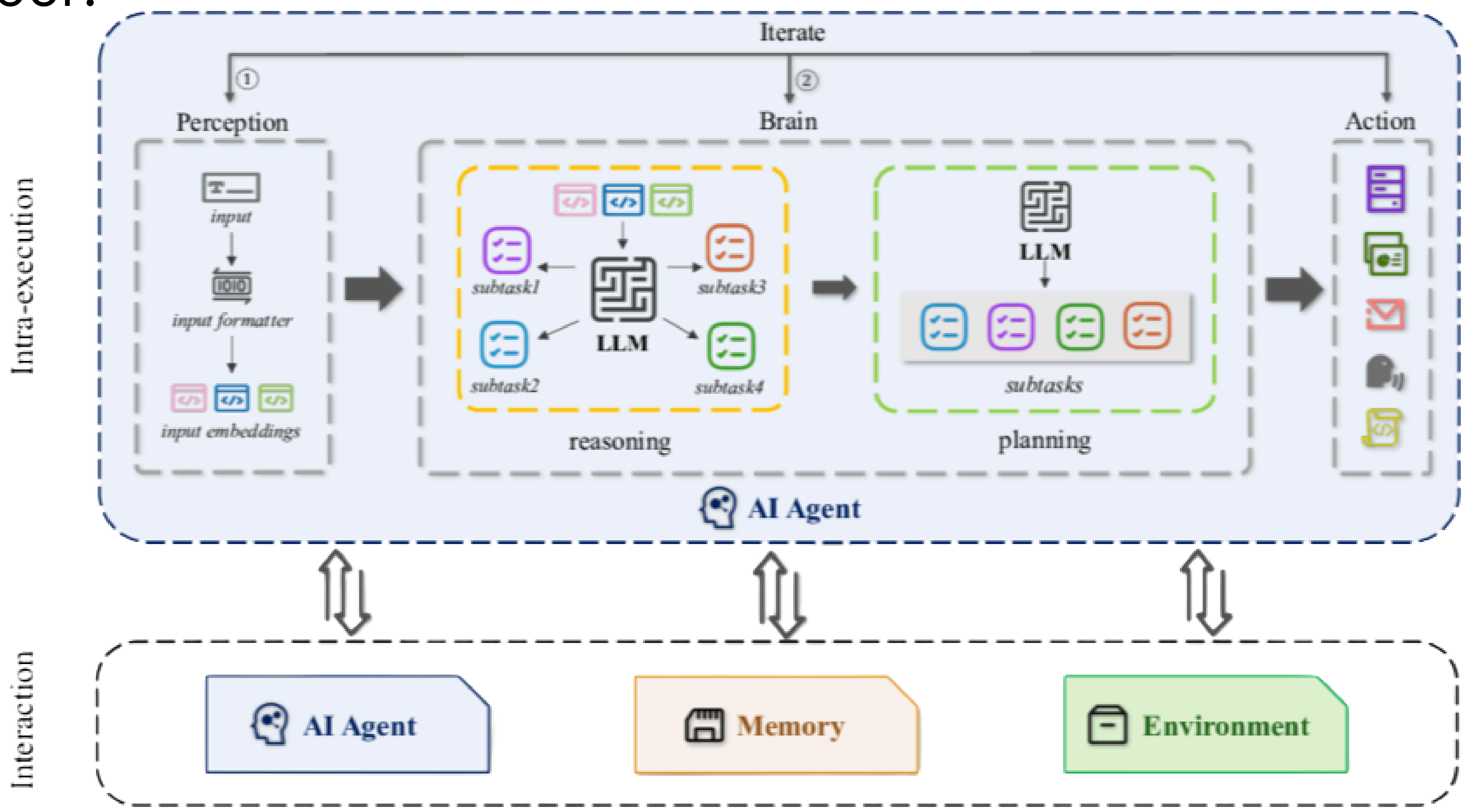
Key Components

- **Text Understanding:** Enables reasoning, planning, and tool use through language models.
- **Visual Perception:** Uses computer vision models to analyze images, video frames, or spatial maps for object recognition, scene understanding, or OCR-based context extraction.
- **Audio Processing:** Converts speech to text (ASR) or interprets non-verbal signals such as tone, pauses, or ambient sounds for context.
- **Video Context:** Combines temporal and spatial data to track motion, actions, or sequences of events critical for robotics, surveillance, and autonomous systems.

Through multimodal fusion, agents integrate these signals into a unified context vector, enabling richer situational awareness and better reasoning accuracy.

Building the Perception of AI Agent

- **Sensing**: Agents perceive their environment through sensing, utilizing sensors such as cameras, microphones, keyboards, or motion detectors. For instance, a camera senses a human hand gesture.
- **Data Collection**: Sensory data is collected and stored for interpretation. This phase may involve preprocessing, feature extraction, and appropriate data representation.
- **Processing**: Agents process the collected data to identify patterns, handle anomalies, and extract relevant information related to their goals. Decision-making processes are then executed based on this processed data.
- **Action**: After processing, agents select actions to achieve their goals. This action selection is guided by the optimal steps toward goal achievement. Actuators, such as speakers, motors, or LEDs, are employed to execute these actions. For example, an automatic door-opening system uses hinge-motors to open a door.



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Analytics Vidhya

Durable State Management for Continuity

Traditional AI models operate statelessly each interaction is independent. In contrast, agentic AI requires durable state management to maintain continuity across sessions and tasks.

Persistent State

Agents maintain an internal representation of ongoing goals, progress, and observations. This state can be stored in memory systems that persist across sessions, allowing the agent to resume tasks or recall relevant context.

Short-Term vs Long-Term Memory

- **Short-Term Memory:** Captures the working context active goals, recent API responses, temporary data structures.
- **Long-Term Memory:** Stores historical context such as user preferences, interaction summaries, and outcomes of previous tasks. This layer typically leverages vector databases for efficient retrieval and similarity-based recall.

State Synchronization

In distributed environments, state management requires synchronization between the agent's internal memory, external databases, and execution logs. This ensures consistency and prevents context drift during long-running or multi-agent workflows.